

TEST 2

ANSWER ALL QUESTIONS

DURATION: 1 HOUR

Write your name, Matrics No. and section on every page of your answer sheets. Scan all your answers and upload the file in .pdf via Blackboard. Make sure you upload the correct file.

QUESTIONS 1 (10 Marks)

- a) A charged particles move in the vicinity of a current-carrying wire as shown in Figure 1. For each charged particle, the arrow indicates the direction of motion of the particle, and the + indicates the sign of the charge. For each of the particles, draw the direction of the magnetic force due to the magnetic field produced by the wire.

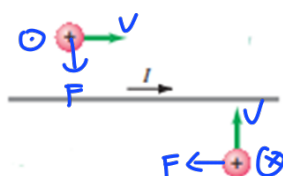


Figure 1

(2 Marks)

- b) An electron moving with a speed of $8.5 \times 10^5 \text{ ms}^{-1}$ in the positive x direction experiences zero magnetic force. When it moves in positive y direction, it experiences a force of $1.5 \times 10^{-13} \text{ N}$ that points in the negative z direction. Determine the direction and magnitude of the magnetic field. **1.10 T, to the left**
(Given $q = 1.60 \times 10^{-19} \text{ C}$ and $m_e = 9.11 \times 10^{-31} \text{ kg}$)

(3 Marks)

- c) A triangular coil STU with $ST = 5 \text{ cm}$, $TU = 4 \text{ cm}$ and $US = 6 \text{ cm}$ having 50 turns of wire is placed in a uniform magnetic field of 0.54 mT as shown in Figure 2. It carries a current of 5.0 A in the direction UTSU. Calculate;

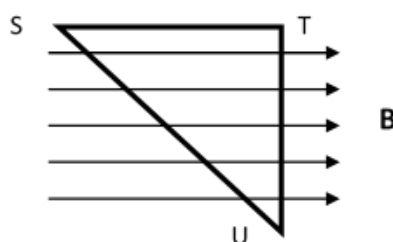


Figure 2

- (i) The magnitude of magnetic force for arm TU. **$5.40 \times 10^{-3} \text{ N}$**

(ii) The magnitude of magnetic force for arm US. $5.06 \times 10^{-3} \text{ N}$

(5 marks)

QUESTIONS 2 (10 Marks)

a) What will happen to the magnetic field when the number of turns in a coil is doubled?

increases

(2 marks)

(b) Figure 3 shows two parallel straight wire, **P** and **Q** are separated at a distance 25 m between each other. The currents in wire **P** and **Q** is 30 A and 25 A respectively when the direction of the current is opposite and the length of each wire is 6 m. Calculate the magnitude and direction of the total magnetic field at the point midway between the wires.

$8.80 \times 10^{-7} \text{ T}$, into the page

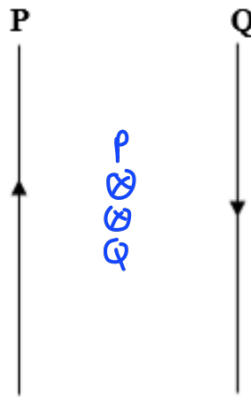


Figure 3

(4 marks)

c) A 54 cm long solenoid with 0.8 cm in diameter is to produce a 0.70 T magnetic field at its center. If the maximum current is 8.7 A. Calculate,

(i) the number of turns must the solenoid have. 3.46×10^4 turns

(2 marks)

(ii) the number of turns per unit length. 6.41×10^4

(2 marks)

QUESTIONS 3 (10 Marks)

- a) State 2 ways to change the magnetic flux through a loop
Change the magnetic field strength (increase, decrease) over the surface area, Change the area of the loop (2 marks)

- b) The self-inductance of a solenoid is given as

$$L = \frac{\mu_0 AN^2}{l}$$

For a solenoid of length 12.0 cm and diameter 3.2 cm has self-inductance of 5.0 mH.

- (i) Find the number of turns in the solenoid 770 turns (2 marks)
- (ii) Calculate the magnitude of magnetic field at the centre of the solenoid if a current of 1.2 A flows in it. $1.16 \times 10^{-3} \text{ T}$ (2 marks)
- c) A generator has a coil of 500 turns with cross-sectional area 20.0 cm^2 . The resistance of the coil is 12Ω . To generate current, the coil is rotated in a magnetic field, 0.2 T at a frequency 50 Hz. Calculate,
- (i) the maximum induced emf. 62.6 V (2 marks)
- (ii) the maximum induced current. 5.2 A (2 marks)

1) (b) $v = 8.5 \times 10^5 \text{ ms}^{-1}$, $F = 0$ at 0° ($\vec{v}$ and $\vec{B}$)
 $F = 1.5 \times 10^{-13}$ at 90° (max $\vec{v} \perp \vec{B}$)

$$F = qvB \sin \theta$$

$$1.5 \times 10^{-13} = 1.6 \times 10^{-19} (8.5 \times 10^5) (B) (1)$$

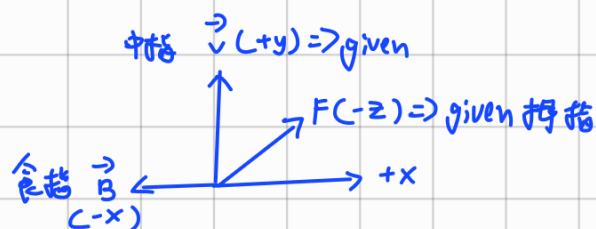
$$B = 1.103 \text{ T to the left (negative x direction)}$$

x direction: $\vec{B}$ (+x / -x)

y direction: $\vec{v}$ (+y)

Using Fleming right hand $\Rightarrow$ (-) charge

z direction: F -z)



(c) $N = 50$, $B = 0.54 \times 10^{-3} \text{ T}$, $I = 5.0 \text{ A}$

(i) $F_{TV} = ILB \sin \theta$ (N)

$$= 5.0 (0.04) (0.54 \times 10^{-3}) (\sin 90^\circ) (50)$$

$$= 0.054 \text{ N}$$

$$\tan \angle TSU = \frac{4}{5}$$

$$\angle TSU = 38.66^\circ$$

(ii) $F_{SV} = ILB \sin \theta$ (N)

$$= 5.0 (0.06) (0.54 \times 10^{-3}) \sin 38.66^\circ (50)$$

$$= 0.00506 \text{ N}$$

2) (a) increase

(b) $B_p = \frac{\mu_0 I}{2\pi r}$
 $= \frac{4\pi \times 10^{-7} (30)}{2\pi (12.5)}$
 $= 4.8 \times 10^{-7} \text{ T}$
 (into page)

$B_Q = \frac{\mu_0 I}{2\pi r}$
 $= \frac{4\pi \times 10^{-7} (25)}{2\pi (12.5)}$
 $= 4 \times 10^{-7} \text{ T}$
 (into page)

$\vec{B} = \vec{B}_p + \vec{B}_Q$
 $= 4.8 \times 10^{-7} + 4 \times 10^{-7}$
 $= 8.8 \times 10^{-7} \text{ T into the page}$

(c) (i) $B = \frac{\mu_0 N I}{r}$
 $0.70 = \frac{4 \times 3.142 \times 10^{-7} \times N \times 8.7}{0.54}$

$$N = 34570.5 \text{ turns}$$

$$\approx 34571 \text{ turns}$$

(ii) $n = \frac{N}{r}$
 $= \frac{34571}{0.54}$
 $= 64019.5$

$$\approx 64020 \text{ turns per meter}$$

3) (a) - change in magnetic field strength

- change in orientation angle b/w $\vec{B}$ and $\vec{A}$

- change in loop area

(b) (i) $L = \frac{\mu_0 AN^2}{l}$

$$5 \times 10^{-3} = \frac{4 \times 10^{-7} \times \pi \times \pi \times (1.6 \times 10^{-2})^2 \times N^2}{0.12}$$

$$N^2 = 593678$$

$$N = 770.5 \approx 770 \text{ turns} *$$

(ii) $B = \frac{\mu_0 NI}{l}$

$$= \frac{4 \times \pi \times 10^{-7} \times 770 \times 1.2}{0.12}$$

$$= 1.16 \times 10^{-3} \text{ T} *$$

(c) (i) $\omega = 2\pi f = 2\pi(50) = 100\pi$

$$\mathcal{E}_{\max} = NBA\omega$$

$$= 500(0.2)(20 \times 10^{-4})(100\pi)$$

$$= 62.84 \text{ V} *$$

(ii) $\mathcal{E} = IR$

$$62.84 = I(12)$$

$$I = 5.237 \text{ A} *$$